



University of
Southern Denmark

MECHANICAL AERIAL SYSTEMS

From VTOL To Fixed Wing: A Two Second Flying Experience

Authors:

Adam Kolodziej	adkol23@student.sdu.dk
Holger J. Jæger	hojae20@student.sdu.dk
Lucas R. Jeppesen	ljepp19@student.sdu.dk
Oliver Torngaard	oltor19@student.sdu.dk

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1 Idea

1.1 Drone functions

The drone should be able to transmit information to people around it, as well as be able to communicate with people in cases of search and rescue operations. The drone should be built as a 2-motor tail sitter. This is to land near people to give the operator a chance to communicate with people during the operation.

1.2 Constraints

The constraints of this drone is as follows:

- Should follow the A2 certification.
- Combined weight of 1 kg.
- Have an easy way to communicate with people.
- Tail sitter / VTOL
- < 1 m diameter length

1.3 Objectives

The drone should be built using lightweight materials, such as foam, 3D print, foil, and should be reinforced with carbon rods in the wings. It should be made to fly for at least more than a couple of minutes, to be able to transmit information, and should be able to safely land in VTOL mode on the tail sitter. The small build should give the drone more maneuverability, and make it easier to reach hard to reach places where potential search and rescue operations could find place.

1.4 Implementations

- Phone/screen/speaker
- Receiver
- Servo motors
- Carbon fiber rods
- 2-3 small size motors
- Tail sitter

2 design

The UAV is a 2 motor tailsitter. Two motors and huge control surfaces are used to be able to easily control the UAV during hovering. The drone is designed in the canard configuration which allows to easily perform tailsitter take off and landing. Vertical stabilizers are also used as landing gear. Original idea is presented in figure 1. Some elements are marked with the colors: carbon parts - red and servos - green. As it can be

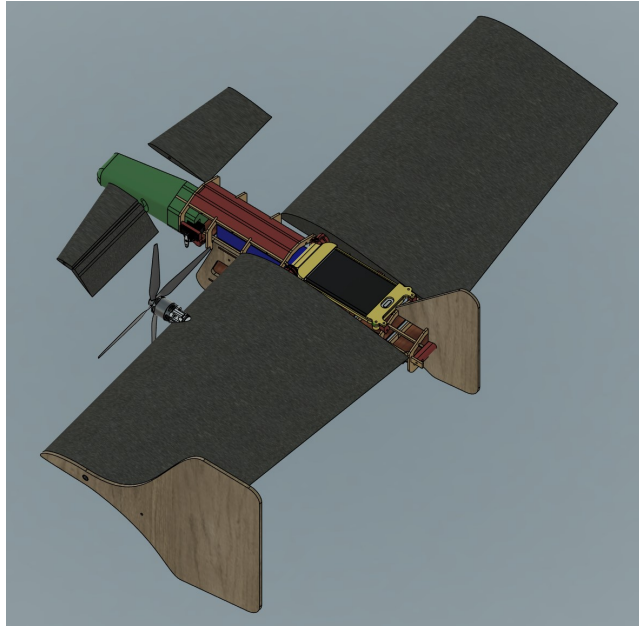
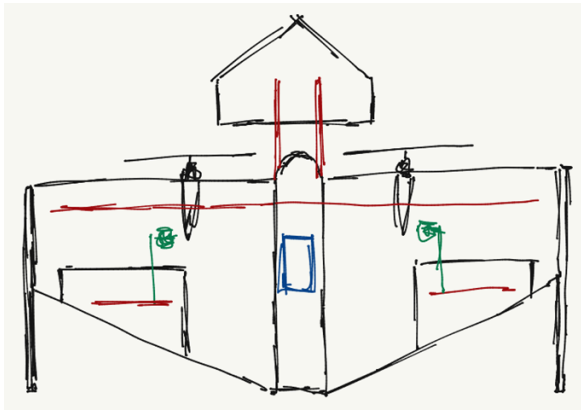
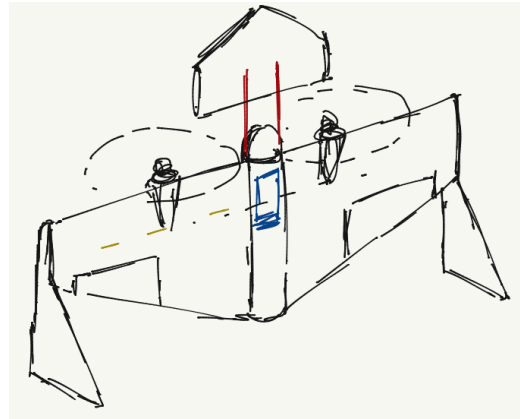


Figure 2: Final design

seen, front horizontal stabilizer was originally passive - it wasn't supposed to have any controlled surfaces. The carbon tubes facing front were also not the safest way to have the canard stabilizer mounted.



(a) 2D sketch



(b) 3D sketch

Figure 1: First sketches

The design evolved to the state presented in the figure 2. The Fusion360 design was done as "half" and not everything was mirrored - that is why there is no second motor or right vertical stabilizer. Due to safety reasons, the nose was done completely with the foam and it became large-volume. Carbon rods were removed. The nose is mounted using wooden plate visible in the figure 3a.

Front horizontal stabilizers are used to control pitch while wings' ailerons are used to control roll. Wings and horizontal stabilizers have carbon tubes and rods going through each pair to make them stiffer ($\phi 10$ and $\phi 4$). Motors are mounted using 3D printed parts and short $\phi 10$ carbon tubes. Everything is secured with M3 screws - figure 3b. In the same picture also mounting system of the main wing rod is shown.

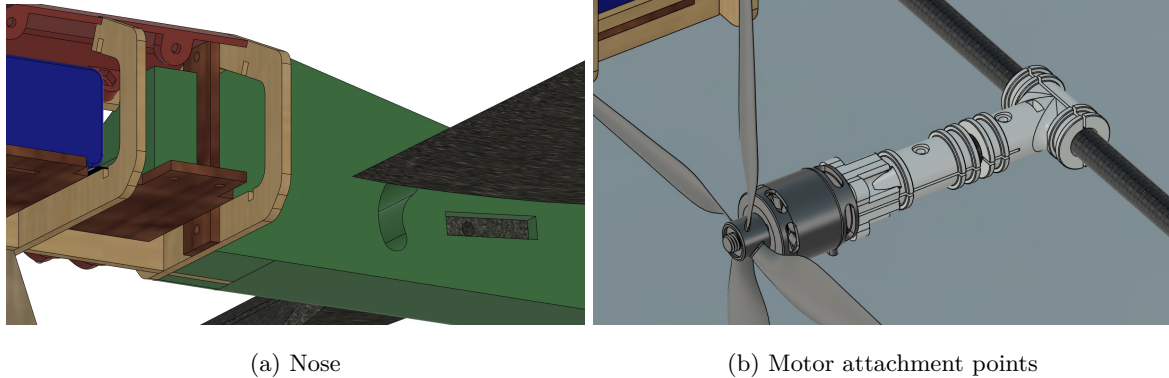


Figure 3: Nose and motor attachment systems

The fuselage is based on balsa and foil to make it stiff but at the same time really light (4). It has two main beams on the bottom. Between them, plates are locked which are used to put electronics and a battery one them. Beams are also connected using balsa ribs. The fuselage was designed to be able to move the battery back and forth to balance the whole UAV. The battery can be strapped down - there are cuts in mentioned plates prepared to pass through straps. Electronics is supposed to be stored under the phone. To provide easy access to it, the phone case is designed to clip on the main $\phi 10$ carbon tube - one the other side it is screwed with two M3 bolts. It is also possible to unscrew then and open the fuselage without taking of phone case completely (5a). The front part has the lid (again screwed with M3 bolts) to have access to battery compartment. As it was mentioned, the vertical stabilizers are also used as landing gear. The UAV has 3 of them (to have required surface needed for roll stability) and the front landing gear. They are designed to not only be used when launching as a tailsitter. The bottom edges aren't parallel to the fuselage. The reason behind that is that we were asked to protect motors and propellers for all cost. The UAV is able to land as normal fixed wing - on belly. Wooden landing gear should provide enough protection for motors 5b.

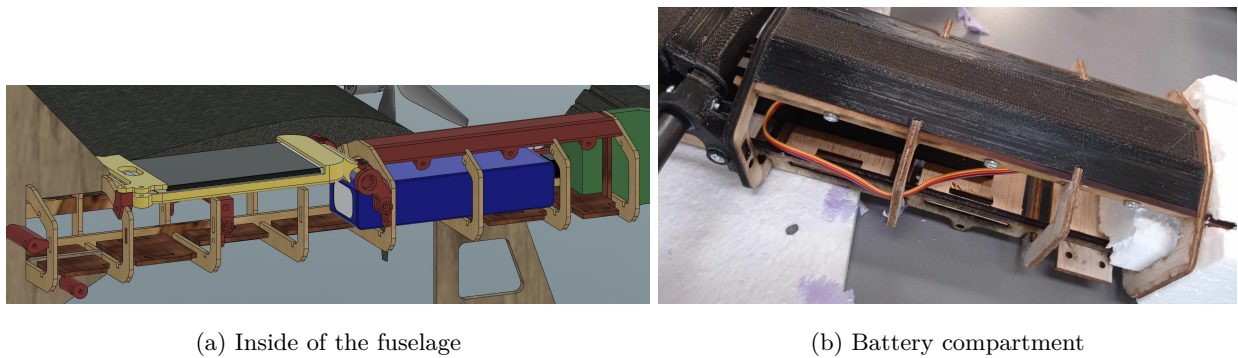
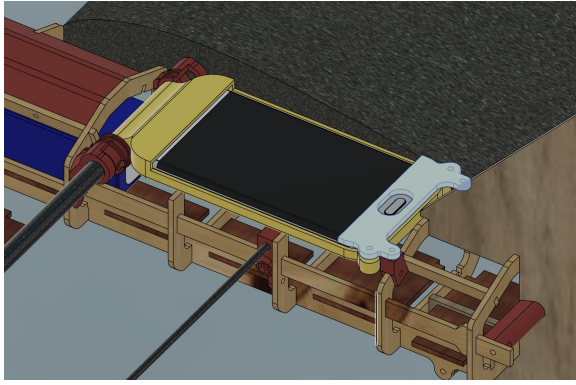
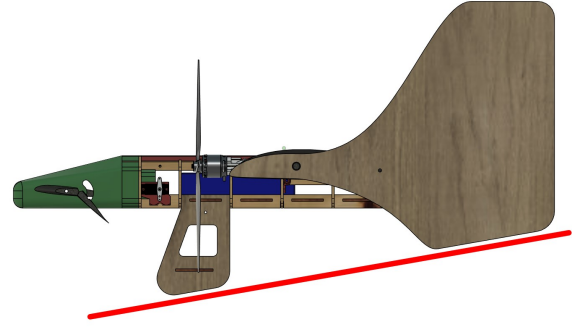


Figure 4: Fuselage



(a) Phone case



(b) Landing gear for traditional belly landing

Figure 5: Fuselage features

2.1 Components - plans vs reality

Ideal configuration with weights:

- EMAX ECO II 2807 1700KV – 47,9g x2 = 100g
- Gemfan 7042 2-blade props or similar – 5,48g x2 = 10g
- 40A ESCs – Skystars Talon 40A – 6,8g x2 = 15g
- Matek flight controller for fixed-wings – eg. F405-WMN – 10g
- Metal gear servo – eg. MG90S – 13,4g x4 = 60g
- 4S1P – 18650 VTC6 LiIon about 3000 mAh – 200g
- Payload – phone – 150g
- Receiver e.g. 20g

Original assumption was to use 2807 motors with 7-inch props which would provide enough thrust and be quite efficient. Moreover, huge thrust would be accessible really rapidly which would increase versatility of the aircraft. In the end, the drone was equipped with motors from IDT (T-MOTOR AIR2216 880KV) with T1045 10" props which had to be taken into account while designing (more prop clearance). The reason is quite simple - using these motors means not spending money on the new 2807. Comparison of the motor performance can be seen in figures 6a and 6b.

Motor type	The voltage (V)	Propeller	current (A)	thrust (G)	power (W)	efficiency (G/W)	speed (RPM)
ECOII-2807-1700KV	16.8	Gemfan7042 2-blade propeller	3.3	300	55.44	5.41	8300
	16.8		6.2	500	104.16	4.80	11000
	16.8		9.8	700	164.64	4.25	12200
	16.8		14.3	900	240.24	3.75	13800
	16.8		19.7	1100	330.96	3.32	15300
	16.8		46.2	1810	776.16	2.33	Max
	16.8						

(a) EMAX ECO II 2807 1700KV

Test Report-AIR2216										
Item No.	Propeller	Throttle	Voltage (V)	Torque (N·m)	Thrust (g)	Current (A)	RPM	Input power (W)	Efficiency (g/W)	Operating Temperature
AIR2216 KV880	T-motor T1045	50%	16	0.07	435	3.5	6015	56	7.77	53.5°C
		55%	16	0.08	527	4.6	6620	73.6	7.16	
		60%	16	0.09	608	5.6	7113	89.6	6.79	
		65%	16	0.11	702	6.8	7563	108.8	6.45	
		75%	16	0.13	888	9.5	8545	152	5.84	
		85%	16	0.15	1076	12.3	9442	196.8	5.47	
		100%	16	0.18	1293	16.2	10464	259.2	4.99	
Notes: Motor temperature is motor surface temperature @100% throttle running 10 mins. (Data above based on benchmark of 2018 are for reference only. Comparison with that of other motor types is not recommended.)										

(b) T-MOTOR AIR2216 880KV

Figure 6: Motors

In connection to motors ([T-MOTOR AIR2216 880KV](#)) from IDT, ESC from the same set were used - AIR20A. It is basically part of the T-MOTOR Air Gear 450 set.

4S LiIon battery would save some weight compared to a 6S pack and have more capacity compared to LiPo but again it was better to just use stuff which was in stock at Drone Center. Last change was not using any flight controller, at least not for the presented project.

2.2 Dual use

The purpose of this drone would mainly be search and rescue operations and for spreading of important information, using the first design with a built-in screen to communicate with people to guide and inform them of current situations. It is important that in search and rescue operations the person feels a connection with the operator, which is the main motivation for a screen. In terms of dual use, the drone could end up being used for evil/bad solutions. An example of this would be the spreading of propaganda since it has a speaker and screen to relay false information to people in its vicinity.

Due to its tailsitter design, the propellers are mounted in front, which could lead to severe damage if colliding with e.g. a person.

Criminals might attempt social engineering attacks by using the drone's communication features to deceive individuals or manipulate them into providing sensitive information or taking actions that could compromise their safety.

3 Aerodynamics

3.1 Wing

To be able to perform the transition from vertical flight to horizontal one (VTOL $\rightarrow$ fixed wing) the drone needs to be able to operate in low velocities and high angles of attack. Ideally, velocity's influence on lift should be as low as possible. We need to also look at the pitching moment which is extremely important during the hovering.

Originally, a 5-degree angle of attack was chosen to get a nice lift and relatively small drag. The calculations for $8m/s$ wind was done. They can be seen in the "value" column. After more consideration, especially how strong the wind can be, it was decided to increase the average speed to $12m/s$. It meant that the lift should have been decreased as well. To achieve that, the angle of attack was changed to 0 degrees. New calculations are presented in the "faster" column 7.

All the calculations and chosen airflow can be found here:

[e334-il\(airfoiltools.com\)](#)

For this VTOL the Reynolds number is rounded up to 200.000 from the calculated 232.280, to fit into the airfoil estimations. With a lift force of 14,79 N, the VTOL will be able to generate enough lift to fly. A pitching moment of -0.05 was chosen since the VTOL needs to be able to make a vertical takeoff.

Parameter	char	unit	value	faster
velocity	V	m/s	8	12
length (chord)	L	m	0,275	0,275
kinematic viscosity of fluid	ν	m^2/s	1,42E-05	1,42E-05
wing area	A	m^2	0,34125	0,34125
fluid density	ρ	kg/m^3	1,204	1,204
Angle of attack	AoA	deg	5	0
lift coefficient	C_L		0,9	0,5
drag coefficient	C_D		0,02	0,02
Reynolds number			154853	232280
Lift force		N	11,832912	14,79114
Drag force		N	0,2629536	0,5916456
Pitching moment	C_M		-0,05	-0,05

Figure 7: Excel sheet used to calculate aerodynamics parameters

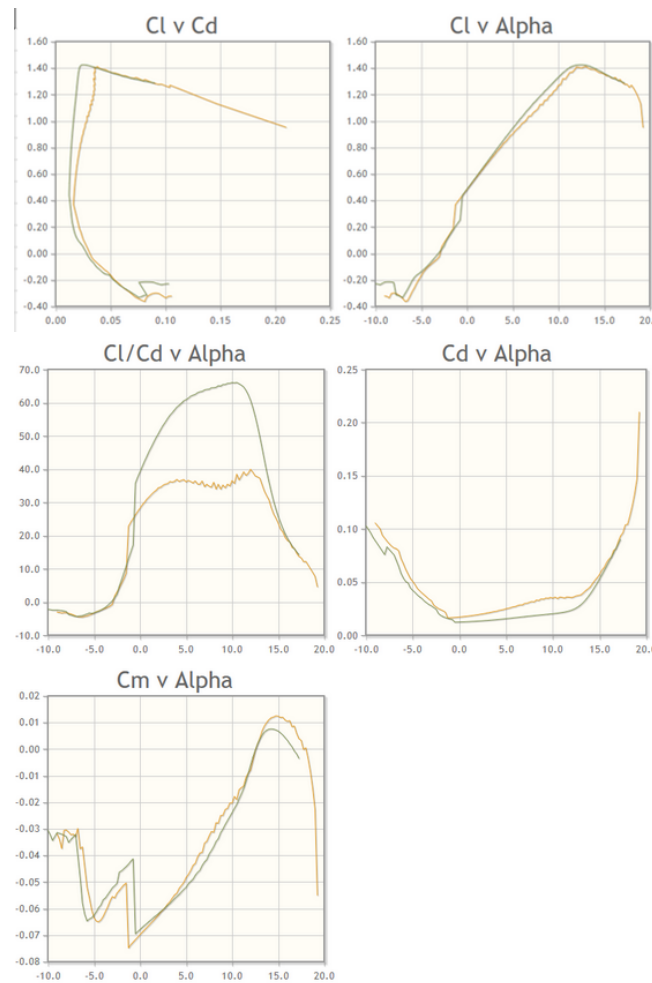


Figure 8: EPPLER 334 AIRFOIL (e334-il) characteristics

Here there is a link to the designed 3D model (imported air flows from airfoiltools.com):

<https://a360.co/3uH3KPw>

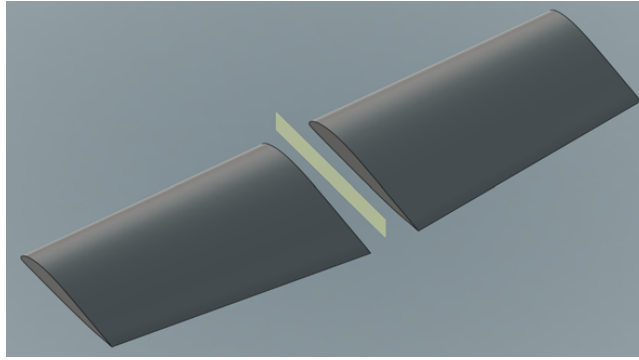


Figure 9: Wings modelled from chosen airfoil

3.2 Power system estimate

The motors draw 275 watts at maximum power, and the system should be able to deliver 550 watts, due to two motors.

General	Model Weight: 1000 g 35.3 oz	# of Motors: 2 (on same Battery)	Wingspan: 1000 mm 39.37 inch	Wing Area: 34.125 dm² 528.9 in²	Drag: simplified 0.03 Cd	Field Elevation: 500 m ASL 1640 ft ASL	Air Temperature: 25 °C 77 °F	Pressure (QNH): 1013 hPa 29.91 inHg
Battery Cell	Type (Cont. / max. C) - charge state: LiPo 3300mAh - 80/120C - normal	Configuration: 4 S 1 P	Cell Capacity: 3300 mAh 3300 mAh total	max. discharge: 85%	Resistance: 0.0039 Ohm	Voltage: 3.7 V	C-Rate: 80 C cont. 120 C max	Weight: 94 g 3.3 oz
Controller	Type - Timing: max 40A - normal	Current: 40 A cont. 40 A max	Resistance: 0.006 Ohm	Weight: 50 g 1.8 oz	Battery extension Wire: AWG10=5.27mm²	Length: 0 mm 0 inch	Motor extension Wire: AWG10=5.27mm²	Length: 0 mm 0 inch
Motor	Manufacturer - Type (Kv) - Cooling: (p = discontinued) T-Motor - Air 2216H-880 (Air Gear 450H) (880) - excellent	KV (w/o torque): 880 rpm/V	no-load Current: 0.8 A @ 10 V	Limit (up to 15s): 275 W	Resistance: 0.115 Ohm	Case Length: 31 mm 1.22 inch	# mag. Poles: 10	Weight: 64 g 2.3 oz
Propeller	Type - yoke twist: T-Motor CF - 0°	Diameter: 10 inch 254 mm	Pitch: 4.5 inch 114.3 mm	# Blades: 2	PConst / TConst: 1.15 / 1.0	Gear Ratio: 1 : 1	Flight Speed: 10 km/h 6.2 mph	calculate

Load: 11.2	Mixed Flight Time: 13.7	electric Power: 261	est. Temperature: 38	Thrust-Weight: 2.40	Pitch Speed: 68

Remarks:	Battery	Motor @ Optimum Efficiency	Motor @ Maximum	Propeller	Total Drive	Airplane
	Load: 11.21 C	Current: 10.13 A	Current: 18.49 A	Static Thrust: 1201 g	Drive Weight: 664 g	All-up Weight: 1000 g
	Voltage: 14.22 V	Voltage: 14.42 V	Voltage: 14.11 V	42.4 oz	23.4 oz	35.3 oz
	Rated Voltage: 14.80 V	Revolutions*: 11167 rpm	Revolutions*: 9958 rpm	Revolutions*: 9958 rpm	Power-Weight: 547 W/kg	Wing Load: 29.3 g/dm²
	Energy: 48.84 Wh	electric Power: 146.2 W	electric Power: 261.0 W	Stall Thrust: - g	248 W/lb	9.6 oz/ft²
	Total Capacity: 3300 mAh	mech. Power: 120.3 W	mech. Power: 205.6 W	- oz	Thrust-Weight: 2.40 : 1	Cubic Wing Load: 5.0
	Used Capacity: 2805 mAh	Efficiency: 82.3 %	Efficiency: 78.8 %	avail. Thrust @ 10 km/h: 1123 g	Current @ max: 36.98 A	est. Stall Speed: 26 km/h
	min. Flight Time: 4.6 min		est. Temperature: 38 °C	avail. Thrust @ 6.2 mph: 39.6 oz	P(in) @ max: 547.4 W	16 mph
	Mixed Flight Time: 13.7 min		100 °F	Pitch Speed: 68 km/h	P(out) @ max: 411.2 W	est. Speed (level): 77 km/h
	Weight: 376 g	Wattmeter readings	Thermal Curve	42 mph	Efficiency @ max: 75.1 %	48 mph
	13.3 oz	Current: 36.98 A	3 6 13 20 s	477 km/h	Torque: 0.20 Nm	est. Speed (vertical): 49 km/h
		Voltage: 14.22 V	31 34 37 38 °C	296 mph	0.15 lb/ft	30 mph
		Power: 525.9 W	87 92 98 100 °F	specific Thrust: 4.60 g/W	Climb Capacity: 3701 m	est. rate of climb: 13.6 m/s
				0.16 oz/W	12142 ft	(-85...90°) 2668 ft/min

Figure 10: Caption

Free body diagram

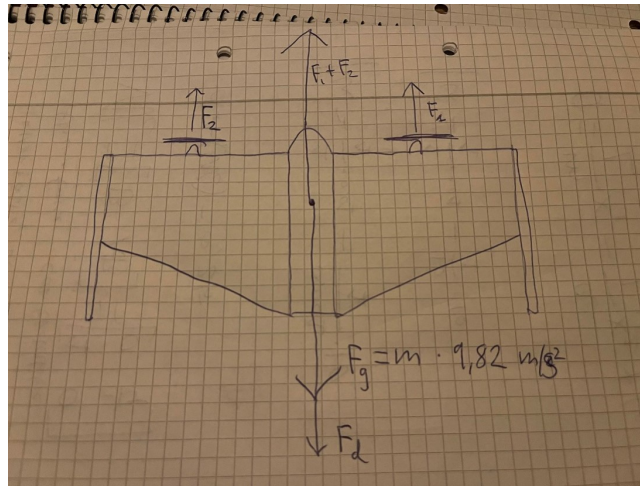


Figure 11: Free body diagram of the drone as seen from its VTOL position. (sitting on the tail)

Assuming vertical lift

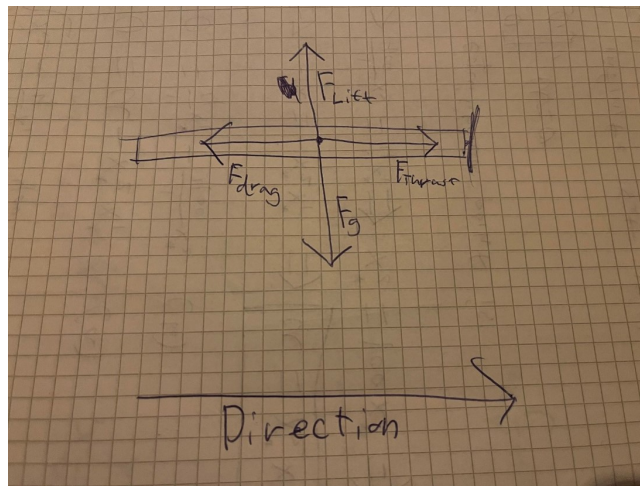


Figure 12: Free body diagram of the drone as seen in a fixed wing flight, where the drone is flying right.

Assuming flying.

4 Stability And Control

To make the drone easier to control we wanna have it aerodynamic stable to decrease the effect of wind, but we do not wanna have too much stability as it will also make it harder for the control surfaces to control the drone

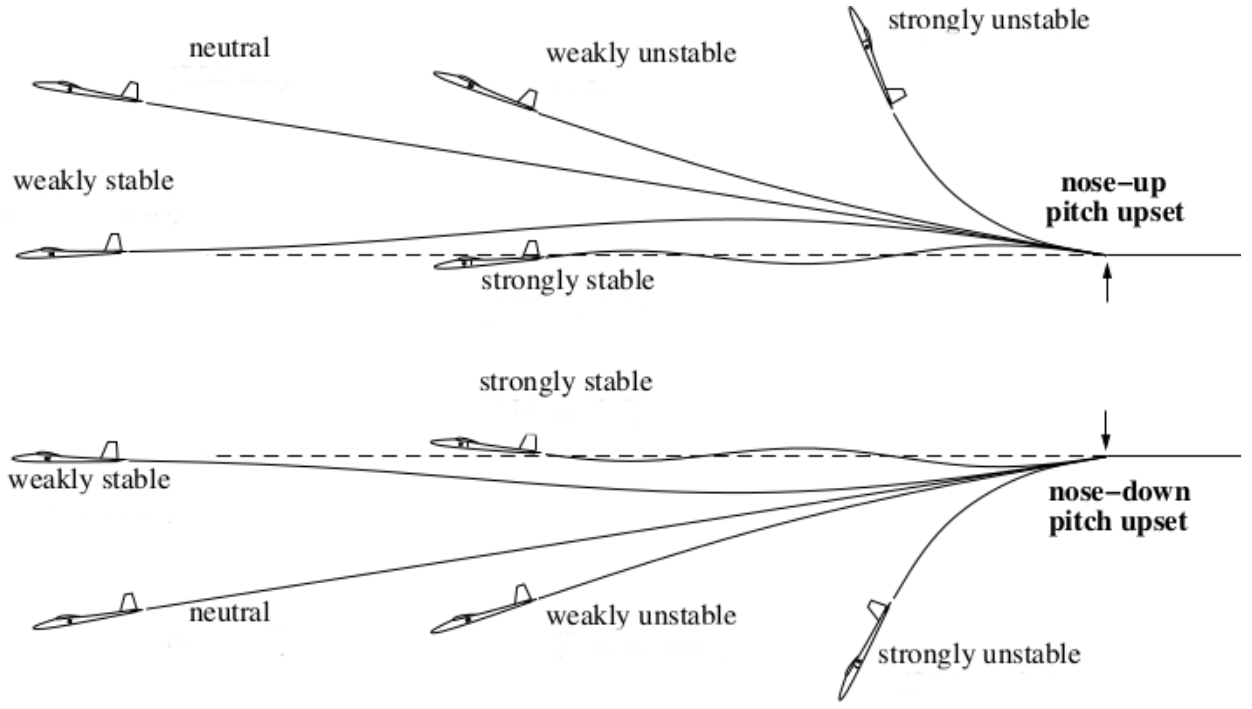


Figure 13: stability in pitch [3]

4.1 Pitch

for stable pitch, the aerodynamic center known as the neutral point is calculated as the center of gravity needs to be in front of this for calculating the neutral point of the drone, the centroid of a flat project of the front half of the wings is used[4]. for calculating the centroid each wing is split into triangles and rectangles as seen in figure 14. to simplify the calculation only the axis along the length of the plane is used as both sides are symmetrical

$$\begin{aligned}
 C_{triangle} &= \frac{h}{3} \\
 A_{triangle} &= \frac{h \cdot b}{2} \\
 C_{rectangle} &= \frac{h}{2} \\
 A_{rectangle} &= h \cdot b
 \end{aligned}$$

where C is a centroid, A is area and h is high for combining the centroids, all the centroid positions are calculated in reference to the front of the drone and then combined using the following formula

$$C_{combined} = \frac{\sum_i C_i A_i}{\sum_i A_i}$$

where C_i is centroid of element i, A_i is area of element i and $C_{combined}$ is the combined centroid

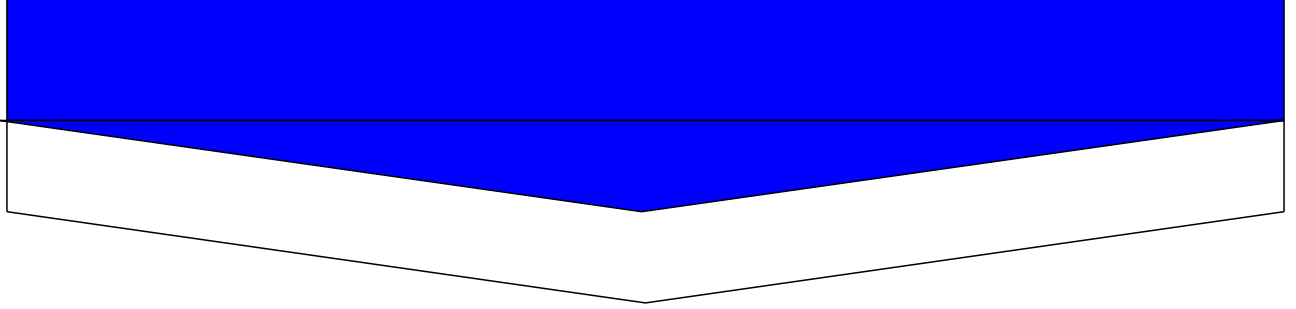
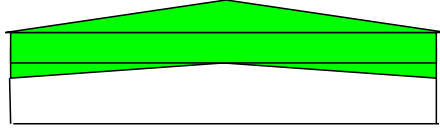


Figure 14: wing profile projection split(illustration)
front wing = green, main wing = blue

after calculation, we get a neutral point of 270.14mm from the front of the drone and given a 10% margin we wanna place the center of gravity at 243.12mm

4.2 Yaw

For yaw stability, we use 3 stabilization tails for yaw damping to help against yaw oscillations. yaw stability is not directly calculated as with sufficient dampening yaw the drone is almost ensured to be yaw stable. As it is less efficient to use multiple smaller tails than one big tail, we assume that the efficiency is 60% of a single tail of the combined area. for finding the yaw damping we wanna have a vertical tail volume coefficient between 0.02 – 0.05 the formula for the vertical tail volume coefficient[3] is

$$V_t = \frac{A_t \cdot l_v}{A_m \cdot b}$$

where A_t and A_m are the area of the tail and main wing area respectively, b is the wing span of the main wing and l_v is the lever arm from the neutral point. the vertical was calculated to be 0.0203

4.3 Roll

as the drone has no dihedral, we have an unstable roll. the roll stability we use the Blaine Rawdon formula [3]

$$B = \frac{l_v}{b} \frac{\Upsilon}{C_L} \quad (\Upsilon \text{ in degrees})$$

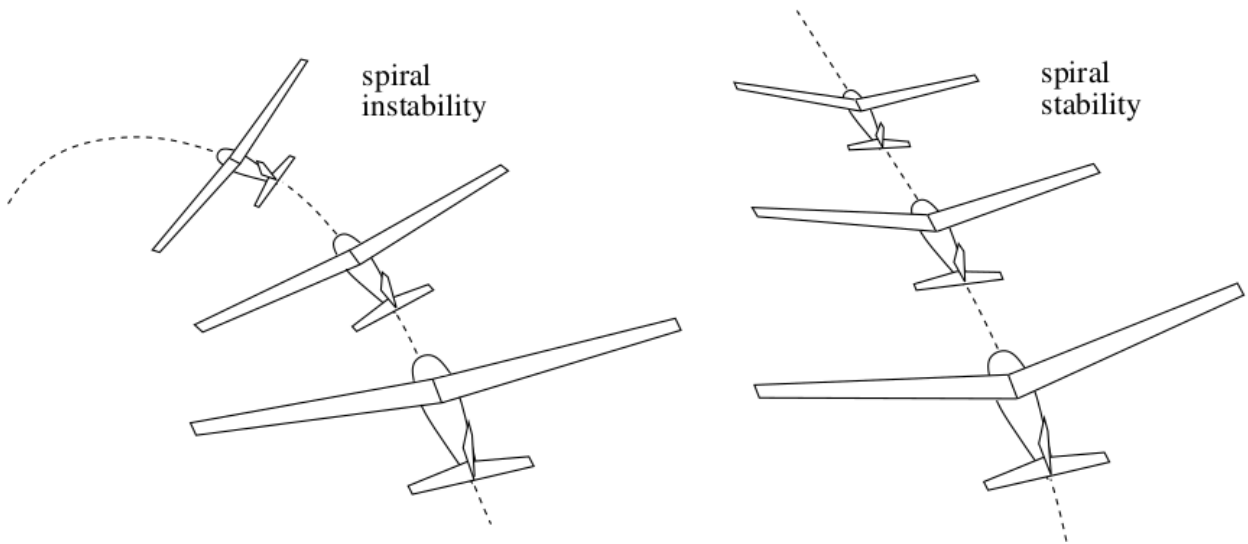


Figure 15: spiral/roll stability [3]

where l_v is the lever arm from the neutral point, b is wing span, Υ is dihedral and C_L is lift coefficient. As there is no dihedral I have an unstable roll, as we need a B value of 5 to be neutral

$B > 5$	, roll stable
$B = 5$	, roll neutral
$B < 5$	, roll unstable

5 Mechanical

5.1 Basic mechanical calculations

Calculating Cantilevered beam/Distributed load - To calculate the distributed load the following formulas were used:

$$M(X) = \frac{1}{2}p(L - x)^2$$

$$M_{max} = M(0) = \frac{L^2 p}{2}$$

$$\sigma_{max} = |M_{max}| \frac{c}{I}$$

By using our initial measurements of the VTOL, the wing length would be 455 mm, while the width would be the mean, being 275 mm. To calculate the line pressure “ p ” the gravity were divided by the load of each wing, being 0.5 kg, which gave 4.91. This would then be multiplied with the mean width resulting in a line pressure load of 1.350 N/m. The Young’s Modulus was found in a lookup table to be 800 MPa, for the chosen material, polyethylene. The moment and maximum bending strength was calculated using the formula for M_{max} giving: 13974 N/m.

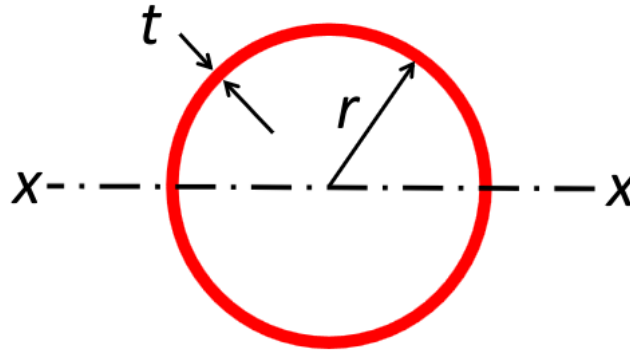
The calculated σ_{max} would be: 17.38 deg.

For the canard wing The line pressure “ p ” the gravity were divided by the load of each wing being 0.1 kg and multiplied with the mean width of the wing, 87.5mm. Giving a p of 0.086 N/m. The moment and maximum

bending strength was calculated using the formula for $M_{\max}$ giving: 966 N/m. The calculated $\sigma_{\max}$ would be: 153 deg.

5.2 Calculating the moment of inertia.

The moment of inertia was calculated from the carbon tubes in the wing. The formulas for this shape is as follows:



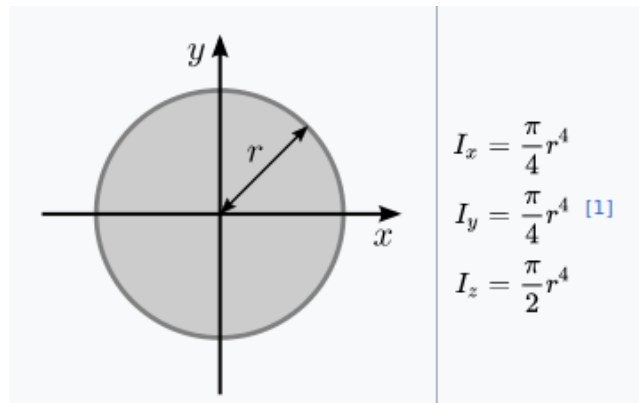
Radius = 4

$t = 2$

hereby:

$$I_{xx} = \pi \cdot r^3 \cdot t = 402.124 \text{ mm}^4$$

For the Canard wing:



Where the moment of inertia was calculated using the formula for I_x Radius = 2

hereby:

$$I_{xx} = \frac{\pi}{4} \cdot r^4 = 12.566 \text{ mm}^4$$

6 Manufacturing

6.1 UAS Manufacturing basic calculations

Determine the manufacturing processes you will use for each component of your drone. Identify processes for the prototype, and for low or high-rate production. Include economic considerations, for example from CES

Edupack Process Universe. Make sure the manufacturing plan matches the context in which the drone would be built in.

6.2 Manufacturing processes and cost

When manufacturing the different components for this drone the methods below were chosen. Carbon fiber – pultrusion: To create the carbon tubes, it was chosen that the pultrusion method was chosen. By using pultrusion to create the carbon fiber tubes, the manufacturing costs are drastically reduced on a small-scale production, compared to the reduction in durability that can be seen in the tubes. On a larger scale production, it could be feasible to do fabric molding, since it would increase the durability of the tubes.

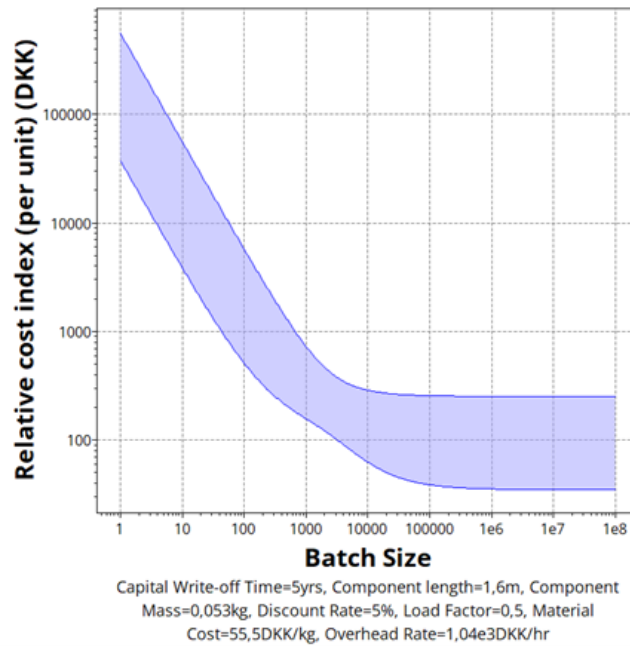


Figure 16: Carbon fiber - price to batch size

6.2.1 Wood (Ply and birch) - Laser cutting

The wood for the drone was processed via laser cutting. Both the birch planks and plywood planks, made for cheap and easily scalable production, since birch and plywood are relatively cheap compared to other composite materials. The only economic consideration would be that the laser cutter is very costly even if it is only a one time buy. For the prototype the birch plywood is handmade, but when scaled it would be bought as pre-fabricated plywood.

6.2.2 PLA – 3D printing, and injection molding.

3D printing represents a cutting-edge digital manufacturing technique renowned for its exceptional precision and cost-effectiveness in producing components. However, this is only utilized in the prototype.

For large scale production purposes, injection molding is preferred, since it is cheaper in a large-scale production perspective, as can be seen in the graph below.

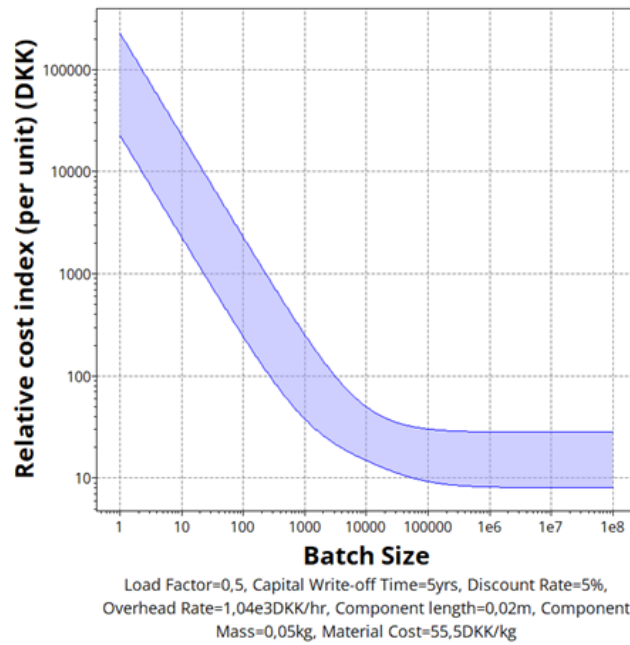
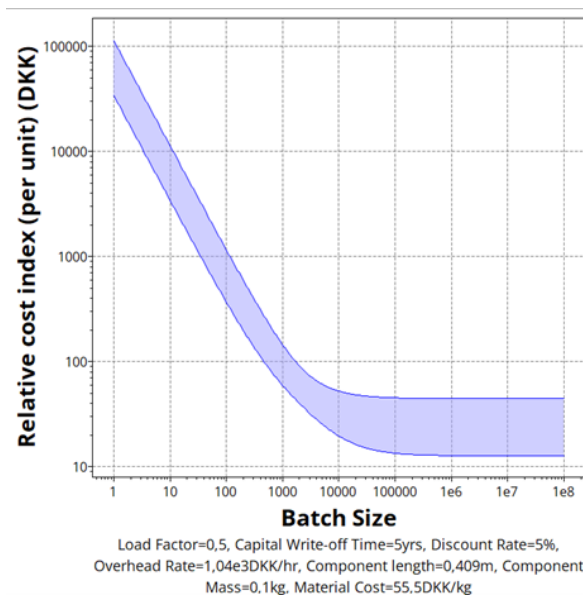


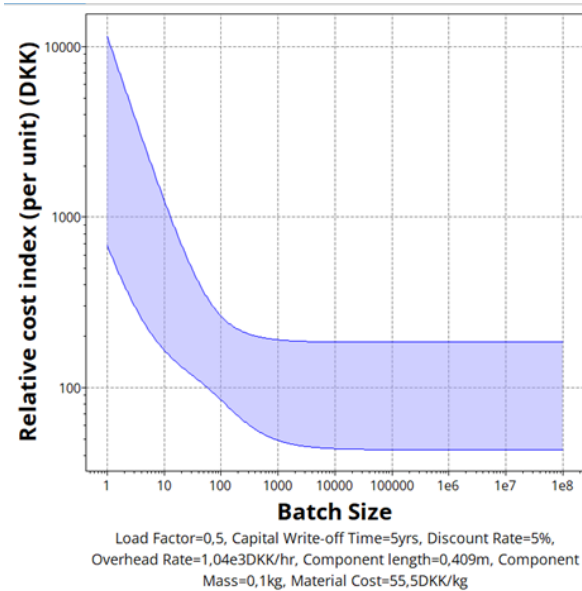
Figure 17: PLA - price to batch size

6.2.3 PET– blow mold / thermo wrapping

PET is used for wrapping of the construction. This can either be done by blow molding or thermo wrapping. For small scale production, thermo wrapping would be ideal since the initial costs are lower, but for large scale production blow molding would be best. For the prototype a hand-made wrapping was done for economic reasons.



(a) Blow mold - price to batch size



(b) Thermo wrap - price to batch size

6.2.4 Rigid foam MD (canard wings) – injection molding / hot wire cutter

The foam for the prototype will be cut using a hot wire cutter, since investing in different molds for the wings would be too expensive for a small-scale production. For a bigger production scale, it would be best to switch

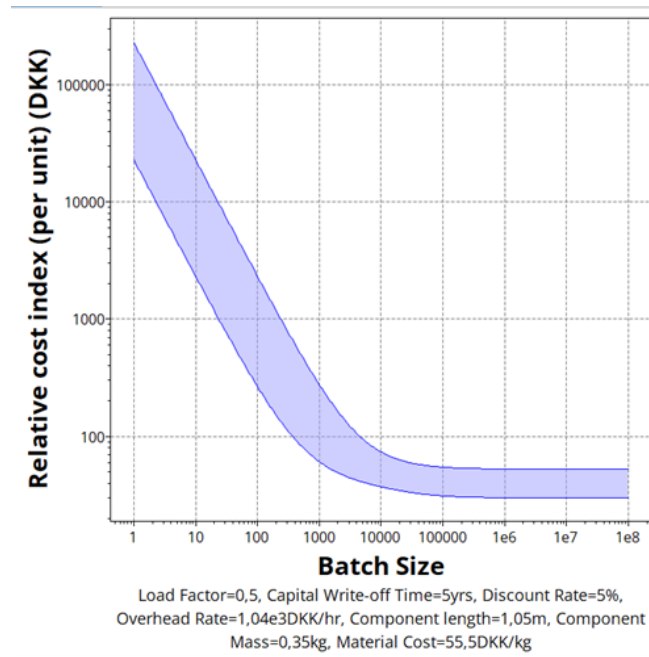


Figure 19: Rigid foam - price to batch size

to injection molding, since production would be sped up, and the production cost would mostly be the same.

6.3 Assembly process

Once all the materials are ready, the assembly process begins.

6.3.1 Fuselage

The fuselage structure primarily consists of balsa wood, with certain pieces being plywood, such as the tail and landing gear.

The balsa wood comes with a thickness of 1 mm. However, a thickness of 3 mm is desired. This is solved by glueing three balsa wood pieces together. By alternating between along grain and across grain of the laser cut balsa wood, we create a 3mm plywood for each “rib” in the fuselage. The balsa wood is cut by utilizing laser cutting, which is a digital manufacturing method. However, a significant drawback of utilizing laser cutting is that cutting must be done before glueing, since laser cutting glue can damage the laser cutter. Even though this slows the manufacturing process down, it is deemed necessary for the health of the laser cutter.

The nose of the plane was cut from foam as well to meet the safety requirements. It couldn’t be cut completely automatically. I required some manual trimming and cuts to be able glue the stabilizers and mount the whole thing on the fuselage. More about the construction can be read in section 2.

6.3.2 Wings and front stabilizers

The wings consist of foam and carbon fiber tube. Foam had to be cut using hot wire. The cut was quite challenging. Two days were spent on that because nobody knew how to cut required shape. Both the wings and the stabilizers had to be cut using *Independent* mode due to two different airfoils on the sides of each part.

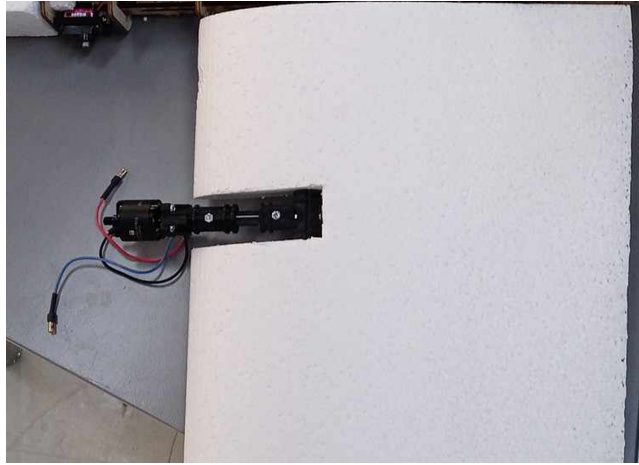


Figure 21: Cut for the motor's mounts

The issues started with the holes for tubes. Several options were tried and tested. Finally wings were cut using two separated procedures. First the holes where cut using simple 2D mode. After that the airfoils were cut using *Independent* mode. Both of the procedures were configured to have the same reference point where the hot wire came back after performing the whole cut. As an example, iXshaper (the program used to operate on the hot wire cutting machine) screens are presented in the figure 20.

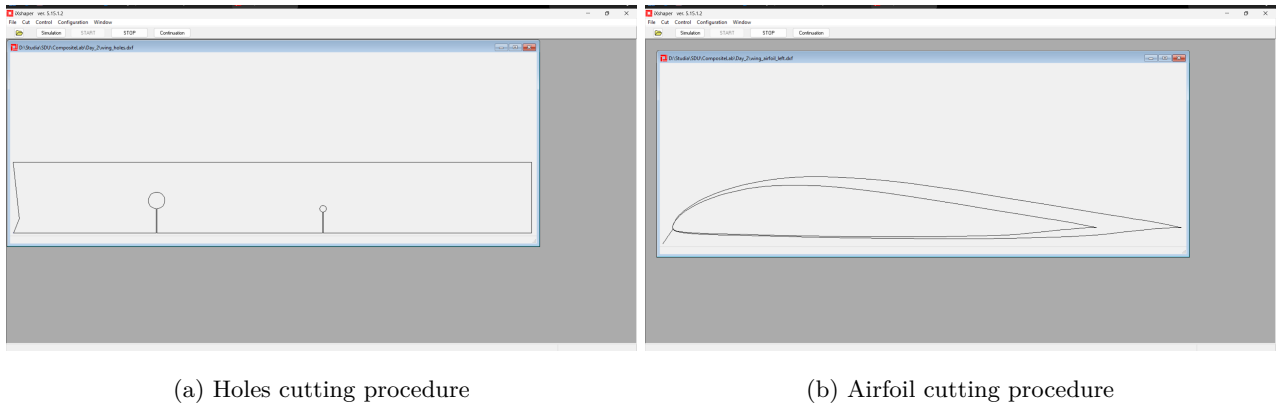
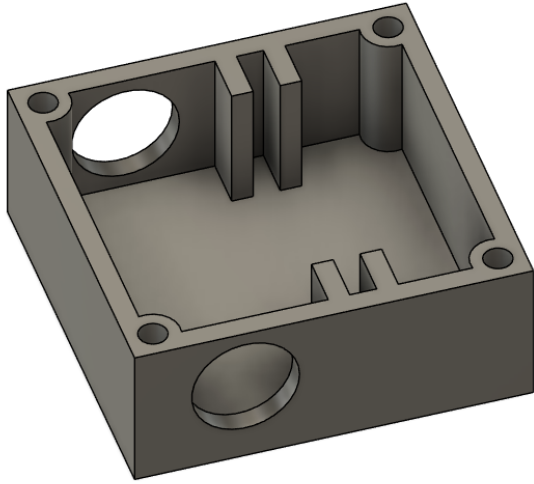


Figure 20: iXshaper screens for the left wing

Stabilizers were cut from MD rigid foam. Again issues were faced. Due to the fact that the stabilizers are much smaller than the wings, they couldn't cool down fast enough - foam was sticking again after the cut. In the end both of the stabilizers were trimmed manually with the knife. Having the wings ready, rectangles were cut to be able to screw motor mounts to the main $\phi 10$ carbon tube. In the picture 21 motor mounted can be seen.

Wings have to be slid on the carbon tube. Then motors on additional tubes should be mounted. The cuts for ESC on bottom side of the wings were done as well as cuts to mount brackets with servos used to control the ailerons 22a. The ailerons are connected via hinges and controlled with servo motors. Hinges weren't so reliable so the connection was straightened with additional pieces of foil 22b



(a) Servos mounting brackets



(b) Aileron

Figure 22: Ailerons control system

Front stabilizers are glued to the nose with additional $\phi 4$ carbon rod to make them stiffer. Process for the control surfaces is the same as with wings.

6.3.3 Front counterweight

It turned out that it's necessary to add some weight to the front because the UAV was back-heavy. It was done using filling free space in the foam nose with M5 screws and nuts. Unfortunately, it wasn't enough. In the picture 23 silver taped on the nose can be seen. It is because huge steel washers were taped on top and bottom of the nose. This way balancing was done.



Figure 23: Counterweight

7 Flight

On May 17th, a test flight was conducted at HCA Airport. As weather conditions plays a significant role in the expectations of a flight, the weather conditions must be stated. The weather was approximately 20 C° with clear skies, providing a great view. However, significant wind conditions were present. The test flight was performed from 14:30 to 15:00. According to data from the National Danish Meteorological Institute (DMI) at 15:00, the maximum wind gust reached 17.8 m/s, and the average wind speed was about 8.9 m/s [2].

Given the weight of the drone, it was initially anticipated that it would perform well under windy conditions. However, due to the strength of the wind, the decision was made to avoid using the VTOL (Vertical Take-Off and Landing) mechanism to prevent the drone from potentially tipping over. Instead, the drone was launched as a fixed-wing aircraft against the wind. This method was chosen because launching against the wind helps generate lift more effectively by allowing the wind to get under the wings with less force.

The takeoff procedure involved one team member controlling the drone and another member preparing to launch it. The motors were powered to maximum speed, and the drone was launched with a gentle but determined force at a relatively horizontal angle from a height of about 1.9 meters. Upon release, the drone initially managed to stay airborne for approximately two seconds. However, it soon went nose-down and crashed, resulting in damage to its nose. The exact cause of the crash remains undetermined. A brief footage of the test flight is available 30 seconds into Dylan Cawthorne's test flight montage [1].

8 Discussion

The very first thing that was considered was the design constraints of the drone, which are defined in section 1.2. One of the constraints which are stated is that the drone must not exceed a combined weight of 1 kg. The reason for this constraint in the first place is first and foremost due to safety concerns. In general, safety and ethics were greatly considered when designing the drone. This specific design constraint proved to be difficult to maintain, as the drone had more back-weight than initially expected. The solution to this issue was to add more weight to the front of the drone. By adding different kinds of bolts and screws near the front part of the drone would account for this. This resulted in a takeoff weight of 2 kg, as the initial drone exceeded the weight constraint aswell. Initially, the exceeded weight was a concern. However, due to the weather conditions described in section 7, the increased weight became advantageous. In strong winds, a heavier drone can be more stable and less susceptible to being a victim of the gusts of wind, providing improved control and flight performance. However, since the total flight time did not exceed two seconds, this is unable to be confirmed in practice.

Furthermore, it was crucial that the tip of the drone did not contain any hard or sharp materials due to safety concerns. To mitigate the risk of injury in the event of a collision with a person, the tip was constructed of foam with a minimum width of 10 cm. This proved to be different than first initially sketched in figure 1, as the pointy edge, as well as the two red sticks in the front proved to be a hazard.

Initially, the drone was designed as a Tailsitter / VTOL, but it was never flown like this. First and foremost due to the weight concerns mentioned, as well as the strong winds that were present on the flight day.

9 Conclusion

In conclusion, the development and testing of the drone showed both challenges and successes in designing a mechanical aerial system. Despite issues like unexpected weight distribution and bad weather, the design was proven to work somewhat in real-world conditions. Adding front counterweights to balance the drone and using foam for safety highlighted the importance of adaptability and safety in drone design.

Initial flights gave useful insights into the stability and control of the drone, showing areas that need improvement. Although the drone's flight time was short, the lessons from these tests will help in making future versions better, focusing on longer flight times and stability in different weather conditions.

Overall, this project has built a strong base for ongoing innovation in drone technology, especially for tasks like search and rescue where reliable performance and safety are crucial. Moving forward, improving the design and using better materials and control systems will be key to creating a more versatile and effective drone.

10 Acknowledgement

The "Extra Extra VTOL"-group would like to thank Dylan Cawthorne for his insightful lectures and his approach to ethical considerations, as well as overall guidance throughout the project.

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